Aaryan S. Maralihalli

Data / Platform Engineer

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Summary

Data/platform engineer focused on **streaming pipelines** — **Kafka + Spark + Redis** event-driven systems. Built a load-balanced real-time pipeline (Locust-tested to **1,000 concurrent users**) and a **1.3B-datapoint** biosignal pipeline (MIMIC-III). Strong Python/SQL. CS graduate (Jul 2026), IEEE-published.

Experience

HyperVerge, Bangalore
Integration Engineer Intern Jan 2026 – Jun 2026

- Monitored **post-go-live data, conversion/drop-off, and stabilization** for enterprise fintech clients; configured rate limits and pipelines for production rollouts.
- Built **"HV Deals Bot"** (Python + Slack) to track and report pipeline/deal state across the team.

Integrated Genetic Solutions (IGS), Bangalore
Project Intern Jun – Aug 2024

- Improved processing of **5M+ blood-lab values** during a dashboard/data migration on a HIPAA-compliant product.

Selected Projects

Live Emostream — event-driven streaming github.com/Aaryansm11/Live_emostream

- **Kafka + Spark Structured Streaming + Redis**, concurrent emoji broadcast over an event-driven architecture; **nginx** load balancing; **Locust** load-tested to **1,000 concurrent users**.

ECG/PPG biosignal pipeline (capstone) github.com/Aaryansm11/ecg-ppg-arrhythmia-prediction

- Built ingestion/preprocessing over **MIMIC-III** — **133K** 10-second segments, 8 channels, ~**1.3B** data points stored in **HDF5**; bandpass filtering, normalization, segmentation.

SDIS github.com/Aaryansm11/SDIS

- Ingestion → chunking → embeddings (FAISS) → **PostgreSQL** with audit logging; FastAPI service, Dockerized.

FinFusion

- Multimodal data pipeline fusing market time-series with news-sentiment features.

Skills

- **Streaming/Data:** Kafka, Spark, Redis, ETL/pipelines, Pandas, NumPy, HDF5, FAISS
- **Storage:** PostgreSQL, MySQL, Neo4j
- **Languages/Infra:** Python, SQL, Java, Docker, AWS (EC2/S3/IAM/ECS), nginx, GitHub Actions (CI), Git
- **CS foundations:** Data Structures & Algorithms (150+ LeetCode)

Publication & Education

- **IEEE ICASI 2026 (Kyoto), Oral Presentation** — *"An Interpretable Model for Stroke Prediction and Arrhythmia Classification using ECG & PPG."*
- **B.Tech, Computer Science** — PES University, Bangalore — 2022 – Jul 2026